Arista Access Node KR-ACC-82: Routing Protocol Flap Troubleshooting and Resolution SOP

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1325

Description

This document outlines the troubleshooting and resolution steps for a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on the Arista Access Node KR-ACC-82, located at KR Puram Access. The failure is characterized by the Alarm Code ALARM_1325 with a severity of Warning. The underlying cause of this failure is due to the complex interplay of routing protocols (BGP, OSPF, and ISIS) and the dynamic nature of the access node's role in the network. The access node's role in the network is to provide connectivity to the core network, and any disruption in this connectivity can cause routing protocol flapping. The Arista equipment, being a software-driven platform, relies heavily on the efficient operation of its routing protocols to maintain network stability. Any misconfiguration, hardware failure, or software issue can cause the routing protocols to flail, resulting in the observed failure. To troubleshoot this issue, it is essential to understand the routing protocol behavior, the access node's configuration, and the network topology. This document provides a step-by-step guide to diagnose and resolve the issue, taking into account the vendor-specific CLI dialect of Arista.

Pre-requisite Checks

- item: Dispatch remote hands to KR Puram Access to assess the site conditions and ensure that all necessary equipment and tools are available., description: Verify that the site is secure and that all necessary safety precautions have been taken before proceeding with troubleshooting.

- item: Check the redundant link status to ensure that the access node is configured to use the redundant link in case of a failure., description: Verify that the redundant link is up and that the access node is configured to use it in case of a failure.

Potential Root Causes

Cause	Probability

Diagnostic Commands & Expected Outputs

```
command: show bgp summary, expected_output: BGP table version is 1, local router ID is 192.168.1.1, description: This command displays the BGP table version and the local router ID, which can help identify any issues with the BGP protocol.
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command: show ospf database, expected_output: OSPF database has 10 entries, description: This command displays the OSPF database, which can help identify any issues with the OSPF protocol.

command: show isis database, expected_output: ISIS database has 10 entries, description: This command displays the ISIS database, which can help identify any issues with the ISIS protocol.

command: show interface, expected_output: Interface is up and operational, description: This command displays the interface status, which can help identify any issues with the access node's interconnects.
```

Resolution Steps

Step 1: step: Clear BGP, OSPF, and ISIS protocols and restart them, command: clear bgp *; clear ospf *; clear isis *, description: This command clears the BGP, OSPF, and ISIS protocols and restarts them, which can help resolve any issues with the protocols.

Step 2: step: Verify that the redundant link is up and that the access node is configured to use it, command: show interface, description: This command verifies that the redundant link is up and that the access node is configured to use it.

Step 3: step: Check the hardware and software configuration of the access node, command: show hardware, description: This command displays the hardware and software configuration of the access node, which can help identify any issues with the hardware or software.

Post-Check Verification

```
command: show bgp summary, expected_output: BGP table version is 1, local router ID is 192.168.1.1, description: This command verifies that the BGP protocol is operational.

command: show ospf database, expected_output: OSPF database has 10 entries, description: This command verifies that the OSPF protocol is operational.

command: show isis database, expected_output: ISIS database has 10 entries, description: This command verifies that the ISIS protocol is operational.
```

Roll-back Procedures

- step: Restore the original configuration, command: copy running-config startup-config, description: This command restores the original configuration of the access node.

- step: Reboot the access node, command: reload, description: This command reboots the access node, which can help resolve any issues with the hardware or software.